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MIL-M-24281(SHIPS)
AMENDMENT - 3
10 April 1970
SUPERSEDING 1/
AMENDMENT - 2
28 September 1967

MILITARY SPECIFICATION

MESSAGE PROCESSING AND DISTRIBUTION SYSTEM

(FOR CVA(N)-68)

This amendment forms a part of MIL-M-24281(SHIPS), 30 January 1967.

Page 3

Paragraph 3.1.8, second sentence: Delete and substitute "The MPDS programming shall provide for rapid system recovery from power transients and power interruption specified in MIL-E-16400 with minimum loss of traffic data other than that contained in messages being processed during power failures."

Paragraph 3.1.14, line 3: Between "7500" and "messages", insert "average".

Paragraph 3.1.15.2, line 1: Delete "3.11.15.1" and substitute "3.1.15.1".

Page 4

Paragraph 3.1.17, line 4: Add a comma after "accomplishment".

Paragraph 3.1.21: Delete and substitute:

"3.1.21 Dependence on facilities control system. The MPDS shall be completely dependent on the Facilities Control System (FCS) for the provision of good communications. Indication of communication channel status at the MPDS control console shall be kept to a minimum."

Pages 4 and 5

Paragraph 3.2.1.2: Delete and substitute:

"3.2.1.2 Interprocessor switching shall be provided such that each equipment peripheral to the central processor group can be selectively switched to any processor."

Paragraph 3.2.1.3, line 2: Delete "computers and multiplexers" and substitute "processors".

Paragraph 3.2.1.5, line 1: Between "terminals" and "shall", insert "typically".

Paragraphs 3.2.1.7 and 3.2.1.8: Delete and substitute:

"3.2.1.7 Interconnection facilities for selecting communication channels and user terminals shall be controllable automatically, semiautomatically, and manually.

"3.2.1.8 Provisions shall be made for connecting and switching printing reperforators, page printers, and transmitter distributors for terminating communication channels as backup in the event of malfunction of the automatic system."

Paragraph 3.2.2.1, line 3: Delete: "multiplexer channels" and substitute "the interconnection facilities".

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

FSC 5895

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Pages 5 and 6

Paragraph 3.2.2.2.4: Delete and substitute:

"3.2.2.2.4 Duplicate message search. A search index compiled for each message from the essential heading elements shall be compared with indices of all other messages held in bulk storage to determine if it duplicates the date-time-group and originator of any of them. Indices shall be retained until the messages are purged from bulk storage. If no duplication is found, the processing sequence shall continue automatically. If duplicate indices are detected, the processor shall store them and cause a display or other readout. Communicator action will then be necessary either to drop the incoming message or continue its processing, with or without (optional) making a visual comparison of the messages to confirm the duplication."

Page 6

Paragraph 3.2.2.2.6.3, line 1: Between "a" and "sequence" insert "terminal".

Paragraph 3.2.2.3, line 2: Delete "at peak rates" and substitute "consisting of 7500 average length messages maximum per day".

Paragraph 3.2.2.3.1, first sentence: Delete and substitute "Retrieval of messages in bulk storage shall be accomplished by keyboard entry of the message system serial number or date-time-group and originator."

Page 7

Paragraph 3.2.2.4.3, first sentence: Delete and substitute "The processor shall interpret each addressee entered by the originator and, by means of table look-up in readily changeable tables of call signs and routing indicators from ACP-117, complete the message heading, and select the proper transmitting channel(s)."

Paragraph 3.2.2.5: Delete and substitute:

"3.2.2.5 Message entry/exit."

Paragraph 3.2.2.5.1: In second sentence; delete "multiplexer capability" and substitute "Capability" and in last line delete "3.3.5.3" and substitute "3.3.6.3 and 3.3.6.4".

Paragraph 3.2.2.5.1.1: Delete and substitute:

"3.2.2.5.1.1 Multiplexing shall provide the functions of serial-to parallel conversion, character and word assembly, and block buffering on incoming traffic and the inverse functions for outgoing traffic."

Paragraph 3.2.2.5.1.2: Line 2, delete "directly". Delete the second and third sentences.

Paragraph 3.2.2.5.1.4, line 1: Delete "incoming".

Paragraph 3.2.2.5.2.2., line 3: Delete "tape readers" and substitute "transmitter distributors".

Page 8

Paragraph 3.2.2.6.1, line 2: Delete "medium speed".

Paragraph 3.3.1: Delete and substitute:

"3.3.1 Central processors. The central processor type shall be CP-642B/USQ-20(V) or similar member of the NTDS family of shipboard approved processors."

Paragraph 3.3.2, line 2: Delete "100" and substitute "150".

Paragraph 3.3.3.3, line 2: Delete "3.3.6.1" and substitute "3.3.5.1".

Paragraph 3.3.3.4, line 2: Delete "3.3.6.1" and substitute "3.3.5.1".

Paragraph 3.3.3.5.1(a): Delete "3.3.7" and substitute "3.3.6".

Paragraph 3.3.3.6, line 2: Delete "processor" and substitute "processors".

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Pages 9 and 10

Paragraphs 3.3.5 through 3.3.7.4: Delete and substitute:

"3.3.5 User terminals. The maximum of twenty user terminals shall be capable of both receiving and generating traffic. They may be of two types: (a) CRT/keyboard/printer and (b) card reader/punch. The interfaces shall conform to the low level interface requirements of MIL-STD-188.

"3.3.5.1 CRT/keyboard/printer terminal. The CRT/keyboard/printer terminal shall be designed modularly so as to allow flexible use in various combinations such as the printer either by itself for receipt of messages only, or in conjunction with the keyboard for message preparation. The terminal shall be capable of operating synchronously at MIL-STD-188 rates up to 2400 bauds. Designated user terminals shall be capable of printing both communication and weather upper case symbols. The terminal shall provide local message storage for at least 1440 characters.

"3.3.5.1.1 The CRT unit shall be capable of displaying at least 20 lines of at least 72 characters per line.

"3.3.5.1.2 The keyboard shall conform to MIL-STD-188 either ASCII or ITA#2 (American version) character set and layout.

"3.3.5.1.3 The terminal shall be provided with a positive security device which must be activated before the terminal can be used. The clearance status of this device shall be capable of interrogation by the processor.

"3.3.5.1.4 Indicators shall be provided for existence and precedence of messages awaiting delivery.

"3.3.5.1.5 The CRT/keyboard portion of the FCC user terminal shall fit in a 19-inch cabinet in accordance with MIL-STD-189.

"3.3.5.2 Card terminal. This terminal, if required, shall contain a card reader and a card punch.

"3.3.5.2.1 The card reader shall be capable of reading at least 400 eighty/column cards per minute. The punch shall be capable of punching at least twenty cards per minute.

"3.3.5.2.2 The terminal shall contain storage adequate to hold data equivalent to two full cards.

"3.3.5.2.3 Controls and indicators shall be provided similar to those of the CRT/keyboard/printer terminal.

"3.3.5.2.4 Operating rates of standard teletype rate, 1200 and 2400 bauds, shall be provided.

"3.3.6 Interconnection facilities and interprocessor switching. The interconnection facilities shall consist of automatic, semiautomatic, and manual switching required to interconnect the user terminals, communication channels, reperforators, transmitter distributors, keyboards, page printers, the MPDS control console, and the Facilities Control Console. The interconnection facilities shall also provide the multiplexing functions specified in 3.2.2.5.1.1. The interprocessor switching shall consist of manual switching required to interconnect the processors, bulk storage, line printer, MPDS control console, circuit sensing multiplexer, and interconnection facilities.

"3.3.6.1 Security. All interconnection facilities shall be treated as RED, and no BLACK data, control, or indicator signals, as defined in Bureau of Ships Instruction 05510.33, shall be included. All signals, including control, shall be low level in accordance with MIL-STD-188.

3.3.6.2 Interfaces. All serial interfaces shall conform to the low level interface requirements of MIL-STD-188. All parallel interfaces shall conform to the standard NTDS interface as defined in NAVSHIPS 0967-051-6040.

"3.3.6.3 Rates. Serial data shall be received and distributed at standard teletype data rates and such other data rates as specified in MIL-STD-188 up to 2400 bauds. Parallel data transfer (full or partial word) shall be capable of proceeding at a rate of at least

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Pages 9 and 10 (Cont'd)

50,000 words per second. For serial internal distribution and collection, message data will be transferred to and from the user terminal synchronously. Optionally, such transfers may be bit parallel, character serial, when in the automatic mode.

"3.3.6.4 Capacity. Provisions shall be included for terminating twenty active receive and twenty active transmit serial communication channels, and twenty half-duplex user terminal channels.

"3.3.6.4.1 Communication channel termination capacity shall be limited to a mix of high speed (1200 or 2400 bauds) and low speed (standard teletype rate) channels, the total not to exceed 16,800 bauds.

"3.3.6.4.2 Message distribution and collection channels shall be made up of a mix of high and low-speed channels, the total rate not to exceed 43,350 bauds.

"3.3.6.5 Interconnection capabilities. The following interconnection capabilities shall be provided:

(a) Any communication channel to the processor group (on-line operation).
(b) Any user terminal to the processor group (on-line operation).
(c) Any incoming communication channel to any reperforator or page printer or both (on-line operation or backup operation).
(d) Any outgoing communication channel to any standard teletype rate transmitter distributor (backup operation).
(e) Any incoming communication channel to either of the two CRT/printer operating stations at the MPDS control console (backup operation).
(f) Any outgoing communication channel to either of the two CRT/keyboard operating stations at the MPDS control console (backup operation).
(g) Any standard teletype rate transmitter distributor or either MPDS control console high-speed paper tape reader to any number of user terminals (back-up operation).
(h) Any equipment peripheral to the processor group to any processor.

"3.3.6.6 Switching. Switching shall be as follows:

(a) Automatic switching control signals for the interconnection facilities shall be originated by the central processor; semiautomatic control signals shall be originated at the MPDS control console. For backup operation, manual switching shall be performed at the interconnection facility.
(b) The degree of backup operation will depend on the equipment failure occurring, such as failures in the MPDS control console, the bulk storage, the high-speed printer, or the central processors. The central processors, if possible, should sense such failures and automatically initiate transfer from on-line operation to the appropriate degree of backup operation.
(c) Manual interprocessor switching to permit interchange of processors.

"3.3.6.7 Indications of interconnections. Indications of interconnections shall be provided to the MPDS control console."

Pages 10 and 11

Paragraphs 4.1 through 4.5: Delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

"4.1.1 Equipment quality accurance.

"4.1.1.1 First article inspection. First article inspection shall be in accordance with the preproduction inspection of MIL-E-16400 and shall be as specified by the developing agency in the individual equipment specifications.

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Pages 10 and 11 (Cont'd)

"4.1.1.2 Quality conformance inspection. Quality conformance inspection in accordance with MIL-E-16400 shall be as specified by the procuring agency in the individual equipment specifications.

"4.2 System quality assurance. A system test plan shall be developed by the Naval Electronics Laboratory Center, San Diego, California, in cooperation with other Naval activities responsible for the development of this system, and the CVAN-68 shipbuilder, to thoroughly test all of the system performance functions required by 3.1.12 through 3.3.6.7, including system integration of individual MPDS equipments into the MPDS, and integration of the MPDS with the FCS. This test plan shall establish test objectives and scope, and furnish detailed test requirements such as parameters, conditions, tolerances, configurations, and sequences."

Page 13

Figure 1: Delete and substitute the attached figure.

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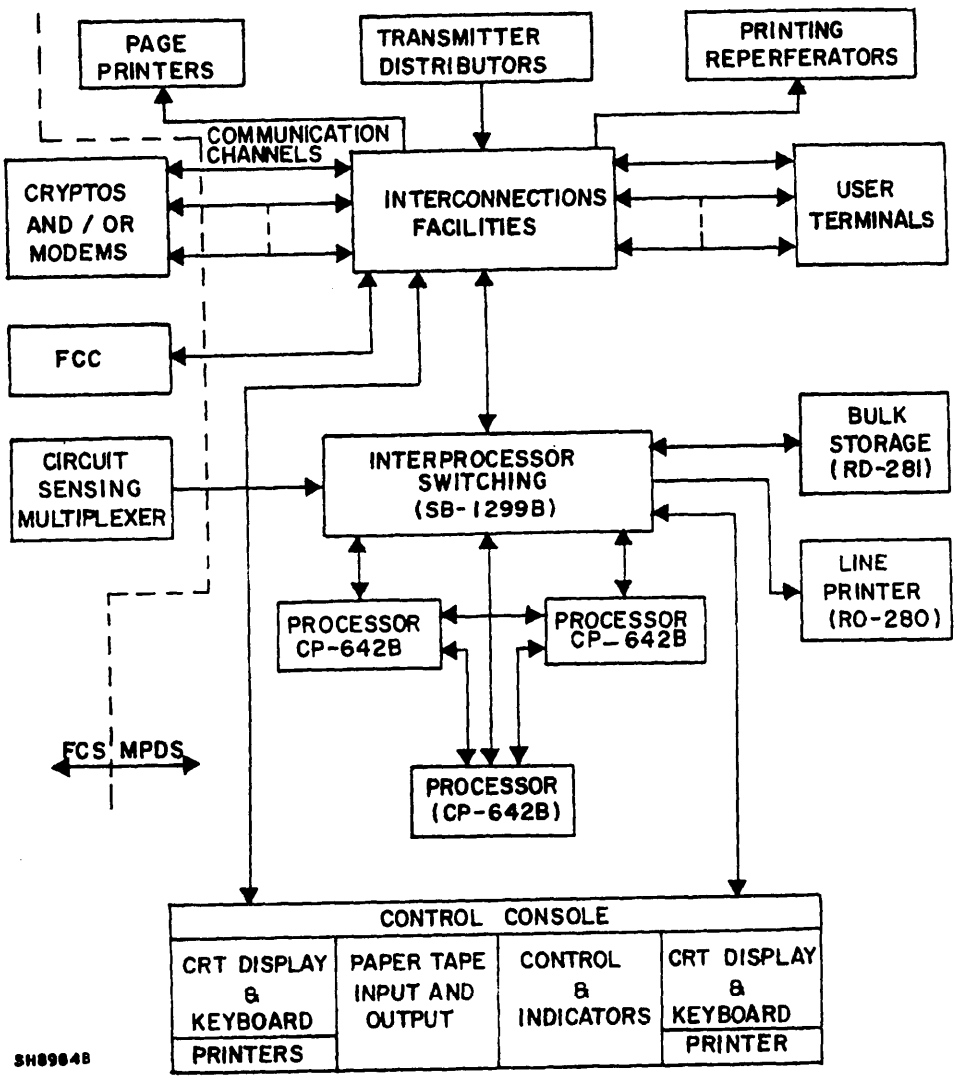


Figure 1 - Block diagram of message processing and distribution system for CVA(N)-68

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